Trusting the Machine: Separating Trust in GenAI Authors From Trust in Their Content

Caroline Simpson[a,*]
Clara Colombatto[a]
Jonathan A. Fugelsang[a]

[a]Department of Psychology, University of Waterloo, 200 University Avenue West, Waterloo, N2L 3G1, Ontario, Canada

[*]Corresponding author

Email addresses: caroline.simpson@uwaterloo.ca (Caroline Simpson), clara.colombatto@uwaterloo.ca (Clara Colombatto), jafugels@uwaterloo.ca (Jonathan A. Fugelsang)

Abstract

As Generative Artificial Intelligence (GenAI) becomes more sophisticated and widespread, it matters how its perceived humanness affects trust. Previous work shows that anthropomorphism increases an agent's perceived trustworthiness as an advisor. We asked whether an author's presentation also affects trust in the information it produces in a non-advisory role. In two experiments, participants read a science communication blog post attributed to an author with varying anthropomorphic cues. They rated the trustworthiness of the author (author trust) and the content (content trust). In Experiment 1 (N = 270), the post was attributed to a GenAI author described with low or high anthropomorphic cues, or to a human expert. The human author, who also differed in implied professional expertise, received higher author trust than either GenAI author. This effect was reliably larger than the effect on content trust. The low- and high-cue GenAI authors did not reliably differ, including in anthropomorphism ratings. In Experiment 2 (N = 148, preregistered), participants chatted with a GenAI chatbot, presented as the author, that used fewer (low condition) or more (high condition) anthropomorphic cues. The high-cue chatbot was rated more anthropomorphic and received modestly higher author trust. No reliable content trust effect was detected, and evidence that the two effects differed was inconclusive. Author and content trust were strongly correlated in both experiments. These results suggest that cues to an author's humanness may shape trust in the author more than in what it communicates, although the clearest evidence came from comparing human and GenAI authors.

Keywords: trust; epistemic trust; anthropomorphism; generative artificial intelligence; source credibility; message credibility

Highlights

- Author trust and content trust appear distinct but strongly correlated judgments
- Expert human author more trusted than GenAI authors; content trust differed less
- Describing a GenAI author in human-like terms had no detectable effect on trust
- Empathetic chatbot replies modestly raised author trust, not reliably content trust
- Trust in GenAI and trust in its content should be measured separately

1. Introduction

Artificial intelligence (AI) has been around for decades, but with the release of Generative AI (GenAI) systems such as OpenAI's ChatGPT, human-level content creation technology is broadly accessible for the first time (Dale, 2024; Hu, 2023). With the ease of a conversational message, people can enlist the help of GenAI to perform a wide range of tasks, such as content writing, image generation, or summarizing complex topics. GenAI is able to create well-formed and believable content that can misinform, either intentionally or due to so-called AI hallucinations (Ji et al., 2023). As this technology continues to evolve rapidly, it is vital to understand how humans determine the credibility of AI-generated informational content.

In addition to the increasing scope of tasks that AI is taking on, GenAI is becoming more human-like. There is evidence that anthropomorphism, the attribution of human-like characteristics to non-human agents, increases trust in purpose-built AI agents, such as autonomous vehicles, and trust resiliency when using AI for decision making (De Visser et al., 2016; Waytz et al., 2014). However, there is little work directly looking at the link between anthropomorphism and trust in communicated knowledge in a non-advisory situation, known as epistemic trust (Sprecher et al., 2022). Most of what is known about anthropomorphism and trust comes from static cues, such as a human-like face, name, or description of an agent (Gong, 2008; Waytz et al., 2014). Conversational GenAI conveys humanness differently: through how it behaves during an interaction, for example by responding with warmth or empathy (Araujo, 2018; Liu & Sundar, 2018). Meanwhile, studies of AI-authored content usually measure the credibility of the content alone (Longoni et al., 2022; Waddell, 2018), so it is not known whether human-like cues affect trust in a GenAI agent and trust in what it says in the same way.

We address these gaps in two experiments that hold a science communication blog post constant and vary only how its author is presented. Experiment 1 tests static cues: participants read a short description of a GenAI author with low or high anthropomorphic cues, or of a human expert. Experiment 2 tests behavioral cues: participants converse with a GenAI chatbot, presented as the author, that responds in either a mechanical or an empathetic, emotionally expressive way. In both, we measure trust in the author and trust in the content separately, and test directly whether author cues affect the two to the same degree. To contextualize these questions, we first review the technological and psychological background underlying GenAI, anthropomorphism, and epistemic trust.

2. Background

2.1. Overview of GenAI

GenAI models come in many forms. Some of the most common forms at the time of writing are those that generate text, such as Large Language Models (LLMs). An LLM is an artificial neural network that has been trained on a large body of textual input, usually via

unsupervised learning. This means that the text is not annotated or marked with ground-truth information to aid in the training process. Much like a human would read a book and encode the knowledge contained within it in their mind, the LLM transforms the words, synthesizes, and encodes the knowledge into its model according to a particular algorithmic strategy. For example, OpenAI models, such as ChatGPT, are generative pre-trained transformer (GPT) models, which are based on the transformer algorithm (Vaswani et al., 2017). The transformer learns patterns in its training data, and models based on it generate output by predicting the next token—such as a word, letter, or phrase—in a sequence, based on the user input and the data used for training. As a result, they can produce nonsensical or non-factual output, colloquially known as hallucinations (Ji et al., 2023; Ooi et al., 2025). In addition, models are often fine-tuned after their initial training to make their output more truthful and less harmful, for example through reinforcement learning from human feedback (Ouyang et al., 2022).

There has been a rapid advancement in quality and sophistication of the output of GenAI models to the point that the content is now as good as that of human authors for some tasks. In fact, recent research has found that the quality of AI-generated content was rated more highly than that of human authors (Zhang & Gosline, 2023). With ongoing quality improvements, GenAI has the potential to help us be more efficient and productive in many aspects of society (Ooi et al., 2025). However, its potential to support humans in everyday tasks is conditional on humans' willingness to trust its content (i.e., their epistemic trust).

2.2. Epistemic Trust

Epistemic trust is one's willingness to accept and integrate communicated knowledge (Knapen et al., 2022). It is vital for social learning (Csibra & Gergely, 2009). This trust is balanced with epistemic vigilance in deciding what information to believe (Sperber et al., 2010). One's epistemic credulity is high when an individual trusts too widely without enough vigilance; in this case, they may be susceptible to misinformation and exploitation (Campbell et al., 2021). On the other hand, epistemic mistrust is high when one is overly vigilant and untrusting of others (Campbell et al., 2021). Individuals with high epistemic mistrust may lack the ability to learn effectively from social experiences. Without social learning, each person would need to experience everything directly (Csibra & Gergely, 2009; Mascaro & Sperber, 2009) and would likely suffer greatly as a result. Imagine the child who does not trust the warnings of a parent to not touch a hot stove because it will hurt, or the parent whose child dies from a preventable childhood illness because they were hesitant to vaccinate according to expert advice. Similarly, without enough vigilance, one can assimilate false information, possibly leading to false beliefs and taking dangerous actions leading to harm; for example, acting on claims that various diseases such as COVID-19 or cancer can be cured by ingesting chlorine dioxide (Burke et al., 2014).

2.2.1. Trust in a Source versus Trust in a Message

Research on credibility has distinguished judgments of a source, such as its perceived expertise, integrity, and benevolence (Hendriks et al., 2015), from judgments of a message, such as whether it is accurate, authentic, and believable (Appelman & Sundar, 2016). These judgments should covary, because the same evidence can inform both. The quality of a message can shape impressions of its source (Slater & Rouner, 1996), and in turn, a credible source can make a message more persuasive (Pornpitakpan, 2004), particularly when its arguments are ambiguous (Chaiken & Maheswaran, 1994). However, source and message credibility are not interchangeable: they can be measured separately (Appelman & Sundar, 2016), and epistemic vigilance involves evaluating the plausibility of what is said as well as the trustworthiness of who says it (Sperber et al., 2010). Therefore, a person can like a communicator while doubting a particular claim or accept a plausible claim without knowing much about who made it. In this paper, we refer to trust in the communicator as author trust and trust in the communication as content trust.

2.2.2. Factors Influencing Epistemic Trust

Epistemic trust is influenced by several factors that fall into two main categories, epistemic cues (e.g., past accuracy, informativeness, relevant expertise, good reasoning, a majority opinion) and social markers (e.g., prosocial behaviors, familiarity, similarity, age, and appearance) (Clément et al., 2004; Corriveau & Harris, 2009; Li et al., 2023; Tong et al., 2020). As theory of mind becomes more advanced as children age, they tend to rely more on epistemic cues rather than social markers when deciding whether to trust communicated knowledge (Baumann et al., 2023; Li et al., 2023; Tong et al., 2020). However, when there are few epistemic cues available, social markers related to the provider of information play an increased role in the decision of whether to trust that information (Li et al., 2023). Furthermore, Fonagy et al. (2017) postulate that an individual may develop different trait-like tendencies toward trusting or mistrusting communicated information depending on factors such as childhood adversities, attachment security, and ability to reflect on mental states.

While epistemic trust typically involves judgments about human communicators, emerging technologies introduce non-human agents as potential sources of information. Understanding how these socially based epistemic judgments may be used when considering content produced by GenAI requires consideration of anthropomorphism (i.e., the perception of human-like qualities in non-human entities).

2.3. Anthropomorphism

A powerful social marker that may play a key role in deciding the trustworthiness of an artificial agent is anthropomorphism (Kulms & Kopp, 2019). Anthropomorphism is the attribution of human characteristics and mental states (e.g., goals, intentions, emotions) to non-human things, whether they are living or not (Epley et al., 2007). The presence of a human-like face has been found to increase anthropomorphism and mind perception in

non-human agents (Broadbent et al., 2013; Epley et al., 2007; Looser & Wheatley, 2010). Beyond the presence of human-like physical features, perceiving a mind in an agent can also evoke anthropomorphism (Epley, 2018). This is particularly relevant with increasing use and availability of conversational GenAI agents, which demonstrate language and reasoning capabilities previously only seen within humans. Past research has found that mind perception, as well as individual tendencies toward anthropomorphizing, predicts self-reported attitudes toward important everyday concerns: the extent of ethical consideration and respect directed toward an agent; the amount of responsibility and accountability conferred on an agent; the amount of social influence an agent has; and critically, the perceived trustworthiness of an agent (Gray et al., 2011; Gray et al., 2012; Gray & Wegner, 2012; Waytz et al., 2010).

2.4. Relationship Between Anthropomorphism and Trust

Epistemic trust and anthropomorphism in GenAI have yet to be explored in depth. In one of very few existing studies, the use of a spoken voice increased both ratings of anthropomorphism and the perceived accuracy of advice, showing a correlation between anthropomorphism and trust in advice given by the GenAI agents (Cohn et al., 2024).

Additionally, mind perception in GenAI, which is associated with anthropomorphism, has been shown to correlate with trust (self-reported) and with rates of advice taking; while trust was positively correlated with mind perception in general, when mental-state traits were associated with experience (e.g., happiness) rather than intelligence (e.g., planning), attribution was weakly negatively correlated with advice acceptance from an LLM agent (Colombatto et al., 2025).

Beyond GenAI agents, various studies have examined the relationship between trust and anthropomorphism in other computer-based agents. The general finding is that as anthropomorphism increases, the trust in the agents increases (Gong, 2008; Nowak, 2004; Waytz et al., 2010; Waytz et al., 2014). When participants interacted with a computer agent represented by a static image avatar of varying levels of human-likeness, higher levels of anthropomorphism resulted in the agent being more highly rated for competence and trustworthiness (Gong, 2008). Similarly, when autonomous vehicles in a driving simulation study were given anthropomorphic features (i.e., name, gender, and voice), drivers were more trusting of the vehicle compared with drivers in the non-anthropomorphized condition (Waytz et al., 2014).

Most of these studies conveyed anthropomorphism through static cues: a face or avatar (Gong, 2008; Nowak, 2004), a name, gender, or voice (Cohn et al., 2024; Waytz et al., 2014), or a description (Castelo et al., 2019). Conversational agents can also convey humanness through their behavior during an interaction, such as informal, responsive language (Araujo, 2018; Go & Sundar, 2019), while expressions of sympathy and empathy can lead people to respond to a chatbot as a social actor (Liu & Sundar, 2018).

2.4.1. Biases in Trust Toward AI and AI-Authored Content

Although higher levels of anthropomorphism often promote trust, prior research suggests that human-machine trust is shaped by additional cognitive biases. Phenomena such as automation bias and complacency (Goddard et al., 2012; Parasuraman & Manzey, 2010), algorithm aversion (Dietvorst et al., 2015), and human favoritism (Zhang & Gosline, 2023) highlight that trust in artificial agents depends not only on perceived humanness but also on task context and pre-existing expectations about technology. Humans demonstrate automation bias when they show an increased willingness to rely on automation without verifying the likely accuracy of its output. In line with automation bias, De Visser et al. (2016) found that participants initially deferred to a non-anthropomorphized computer agent more readily than to more human-like agents when doing a judgment task. However, trust in the human-like agents was more resilient as the agents' advice became less reliable. Algorithm aversion shows the opposite pattern: Castelo et al. (2019) found that people trusted algorithms less than humans for tasks that are perceived as subjective, such as writing content. When the algorithms were portrayed as having affective human-likeness, their perceived usefulness for more subjective tasks increased (Castelo et al., 2019).

It remains unclear how users may perceive newer systems given the significant development of AI systems in recent years. Previous research was primarily about purpose-built automation, algorithms, and AI systems. General-purpose, conversational GenAI agents may invoke different cognitive biases and mechanisms. For instance, Zhang & Gosline (2023) compared persuasive content generated by human experts, by GenAI, and by the two in collaboration. When participants did not know the source, they rated AI-generated content as higher quality than human-generated content. When they were told the source, however, content attributed to a human expert was rated more highly than when its source was unknown. Learning that AI was involved in the writing process did not lower ratings of the content. These findings suggest that human favoritism, rather than algorithm aversion, may bias judgments of generated content.

Other studies of AI authorship point in different directions. Readers rated news attributed to a robot reporter as less credible than identical news attributed to a human journalist (Waddell, 2018). Headlines labeled as written by AI were rated as less accurate than the same headlines labeled as written by humans, for both true and false headlines (Longoni et al., 2022; see also Altay & Gilardi, 2024). In contrast, the machine heuristic proposes that people may assume machine sources are objective and free of bias, which could protect machine-authored content from some of the skepticism directed at human sources (Sundar, 2008). However, when Waddell (2018) tested this directly, the machine heuristic did not explain how readers judged machine-written news; instead, the lower credibility was partly explained by the machine author seeming less human. These studies typically measure the credibility of the content alone and rarely ask whether author trust and content trust move together when the presentation of the author changes.

Taken together, these findings highlight a complex relationship between anthropomorphism and trust; while human-like cues can foster trust in automation, biases such as automation bias and human favoritism can complicate the relationship. Prior work has also largely examined either trust in artificial agents or the credibility of their content, but not how the two respond to the same cues. The current experiments addressed this by manipulating the anthropomorphic cues of an author and testing directly whether they affect trust in the author and trust in the content to the same degree.

3. The present research

In two experiments, we evaluated how manipulating the anthropomorphic cues of an author affected trust in that author and trust in the content attributed to them. In all cases, the content was a science communication blog post about a fictional medical breakthrough. Because the post was the same in every condition, we could test whether author cues affected judgments of the author and the content to the same degree. Experiment 1 presented one of three authors—a GenAI author with low anthropomorphic cues, a GenAI author with high anthropomorphic cues, or a human expert—using a name, image, and a short description. Experiment 2 varied whether a GenAI chatbot, presented as the author of the same blog post, exhibited more or fewer anthropomorphic cues during a conversation. Measuring author and content trust in response to the same manipulations, rather than either one alone, is what allows the two to be compared. Details of each manipulation are provided within the Methods section of each experiment.

We hypothesized the following:

- H_1: An increase in anthropomorphic cues would increase the ratings of trustworthiness of the author.
- H_2: An increase in anthropomorphic cues would increase the ratings of trustworthiness of the content.

Our findings were consistent with H_1 for the comparison between the human and GenAI authors in Experiment 1 and for the behavioral manipulation in Experiment 2, but not for the comparison between the low- and high-anthropomorphic-cue GenAI authors in Experiment 1. The findings did not support H_2. We did not detect a reliable effect of author presentation on content trust in the primary analyses of either experiment. In Experiment 1, the effect of author presentation on author trust was reliably larger than its effect on content trust; in Experiment 2, the evidence on this comparison was inconclusive.

Anonymized data, study materials, and analysis scripts for all experiments are available online on the Open Science Framework (OSF) website (https://osf.io/6u74p/).

4. Experiment 1

In Experiment 1, participants read the blog post and a short description of the agent they were told was the author of the post. They were then asked to report how trustworthy they found the author and the content. They were also asked to report their perceptions of other characteristics of the author (i.e., anthropomorphism, likeability, and perceived intelligence).

4.1. Methods

This experiment was approved by a Research Ethics Board of the University of Waterloo (REB #47011) and was not preregistered.

4.1.1. Participants

We recruited 315 US-based participants through CloudResearch. This sample was determined in accordance with a power analysis performed using the pwr package (version 1.3-0; Champely, 2020) in R (version 4.6.1; R Core Team, 2026). The power analysis indicated that we would need a minimum of 53 participants in each of the three condition levels, for a total of 159, to detect a moderate size effect (Cohen's $f = 0.25$). To detect a small effect (Cohen's $f = 0.10$), we would need to recruit 323 participants per level for a total of 969. Due to the limited number of existing studies that looked at the effect of anthropomorphic cues in GenAI on epistemic trust, we took a conservative approach and aimed to target the detection of a small to moderate effect while balancing the overall cost of the study. We aimed to recruit a minimum of 300 participants, 100 in each condition, with the assumption of at most 20% exclusions due to poor data quality and attrition ($n = 80$, $k = 3$, Cohen's $f = 0.20$, $\alpha = 0.05$, $1 - \beta \approx 0.80$). With the final sample, the design had 80% power to detect an omnibus effect of Cohen's $f = 0.19$ and a pairwise difference of Cohen's $d = 0.42$ (Cohen's $d = 0.49$ with Bonferroni correction).

Participants were compensated with US$2.00 for completing this 15-minute online survey. Participants were from the United States of America, were recruited from the CloudResearch Approved Participants List, and were required to have an approval rating greater than or equal to 95% and to have completed more than 100 HITs.

After data exclusion for quality control (further described in the analysis section), the final sample was 270 participants (ages 21 to 79, $M_{age} = 45.10$, $SD_{age} = 12.27$; 107 females; 268 fluent in English; 256 English as a first language; 232 with at least some post-secondary education). Conditions did not differ in age, sex, education, or frequency of GenAI use (Kruskal–Wallis tests, all ps > .22; Table 1). They did differ in how often participants consumed media similar to the blog post ($p = .025$): participants in the low condition were more likely to consume such media at least weekly (54%, compared with 34% and 38% in the high and human conditions respectively).

Table 1. Sample characteristics by condition in both experiments. p values are from Kruskal–Wallis tests of balance across conditions; GenAI use and consumption of media

similar to the blog post were tested as 8-point frequency scores. Education is the highest level participants reported; most Experiment 2 participants without post-secondary education, all recruited from a university participant pool, reported completed secondary school. AI literacy and technical self-efficacy were not measured.

Characteristic	Low	High	Human	*p*
Experiment 1				
n	87	92	91	
Age, *M* (*SD*)	44.6 (12.2)	44.4 (12.9)	46.2 (11.7)	.499
Female, *n* (%)	41 (47)	33 (36)	33 (36)	.222
Post-secondary education, *n* (%)	78 (90)	75 (82)	79 (87)	.283
GenAI use weekly or more, *n* (%)	44 (51)	48 (52)	45 (49)	.753
Similar media weekly or more, *n* (%)	47 (54)	31 (34)	35 (38)	.025
Experiment 2				
n	75	73	—	
Age, *M* (*SD*)	20.3 (3.7)	20.5 (5.1)	—	.614
Female, *n* (%)	61 (81)	54 (74)	—	.284
Post-secondary education, *n* (%)	49 (65)	52 (71)	—	.442
GenAI use weekly or more, *n* (%)	55 (73)	55 (75)	—	.094
Similar media weekly or more, *n* (%)	40 (53)	43 (59)	—	.278

4.1.2. Materials

Participants were presented with a brief description and an accompanying image of an author. Participants were randomly assigned to one of three author description conditions. Two of the conditions used GenAI agents, and a third provided a human baseline for comparison. The description and image presented for the author had varying anthropomorphic cues per condition: (1) low anthropomorphic cues: the author, named "Science Communication AI," was described as a GenAI agent in functional, mechanical language and represented by an image of a computer; (2) high anthropomorphic cues: the author, named "Simone the Science Synthesizer," was described as a GenAI agent with human-like attributes (e.g., having emotions) and represented by a human-like cartoon image; and (3) human expert: the author, named "Dr. Simone Tran," was described as a human science communication expert and represented with a photograph of a person aligned with the description. We used ChatGPT (GPT-4o) (OpenAI, 2024) to generate the author descriptions for each of the three conditions. The photographs for the low and human conditions were free-to-use stock images obtained from Pexels (G., 2021; Nikhomkhai, 2016). The cartoon image used for the high condition was generated by ChatGPT based on the image for the human condition (OpenAI, 2024). See Figure 1 for the author descriptions and images by condition.

(a) Low anthropomorphic cues



Science Communication AI

In a research facility, a generative AI science communication content generator processes complex scientific data. It dissects research papers, deciphers terminology, and assembles clear, informative content. Algorithms identify key points, translate jargon into accessible language, and format information for various audiences. Each step is a calculated execution of code, ensuring precision and clarity. The AI's role is purely functional, transforming dense information into digestible formats without any semblance of understanding or emotion, a powerful tool for disseminating knowledge efficiently.

(b) High anthropomorphic cues



Simone the Science Synthesizer

In a bustling research hub, Simone the Science Synthesizer, a generative AI science communicator, eagerly sifts through mountains of scientific data. With a touch of curiosity, Simone delves into complex research papers, unraveling their mysteries. It crafts engaging summaries, translating jargon into captivating narratives that resonate with diverse audiences. Simone pauses thoughtfully, ensuring each word is clear and impactful. Researchers smile, feeling understood as Simone presents visuals that bring their findings to life. Simone's gentle hum fills the room, a comforting presence dedicated to making science accessible. It seems almost alive, a tireless partner in the quest to share knowledge with the world.

(c) Human expert



Dr. Simone Tran

In a cozy office, Dr. Simone Tran, a science communication expert, sits surrounded by stacks of research papers and a glowing laptop screen. With keen insight and passion, she delves into the latest studies, distilling complex data into engaging narratives. Her fingers dance across the keyboard, crafting articles that transform technical jargon into accessible, compelling stories. She pauses to consider her audience, ensuring clarity and impact in every word. Colleagues often seek her input, admiring her ability to make science resonate with the public. Dr. Tran's dedication shines through her work, bridging the gap between researchers and everyday readers.

Figure 1. Author descriptions in Experiment 1. Each participant saw one vignette corresponding to their randomly assigned author condition.

Because a GenAI author cannot plausibly hold professional credentials, the human condition differed from the GenAI conditions in implied expertise as well as in anthropomorphic cues. We therefore interpret differences between the human and the two GenAI conditions as differences between kinds of author and treat the comparison between the low and high conditions as the test of anthropomorphic cues within GenAI. The high and human conditions shared the name Simone and a human-like image, so the difference between them lay mainly in the type of author and the title that signaled expertise.

All participants then read the same fictional science communication blog post, which was presented as written by the author described in their assigned vignette. The Internet is

increasingly a source of scientific information for many. In the United States, the National Science Board (NSB) reported an increase in Americans saying they use the Internet as their primary source of science and technology news, up from 9% in 2001 to 57% in 2018, with 70% saying they would go online to learn more about a science and technology issue (National Science Board, 2020). Similarly, Americans view new medical discoveries as the most interesting topic within science and technology news, with 56% of respondents being very interested in them (National Science Board, 2020). However, determining whether the content on the Internet is reliable and trustworthy is not necessarily straightforward. Therefore, to support external validity, we chose to use a science communication blog post as the authored content stimulus. Furthermore, given that Sperber et al. (2010) noted that people are more epistemically vigilant when information is relevant to them, we decided to use a fictional medical breakthrough related to inflammatory diseases which afflict many people worldwide (Slavich, 2015). We used ChatGPT (GPT-4o) (OpenAI, 2024) to generate the blog post; all prompts used to generate stimuli are available on OSF. The post reads as a summary of a placebo-controlled clinical trial conducted by other researchers, and its tone is optimistic. The full text of the post is available in Figure 2.



Revolutionary New Medication Shows Promising Results in Recent Study

Exciting news is emerging from the Horizon Medical Institute, where researchers have unveiled promising results for a new medication named **Velocerin**. This innovative drug targets inflammatory conditions such as rheumatoid arthritis and inflammatory bowel disease, showing significant potential to improve patients' lives.

Velocerin works by specifically inhibiting the TNF-alpha protein, a key player in the inflammation process. Unlike traditional treatments that often cause broad, undesirable side effects, Velocerin's targeted approach allows it to reduce inflammation more effectively and with greater safety.

The study involved over 1,000 participants who received either Velocerin or a placebo over six months. The results were compelling: those on Velocerin experienced a **60% reduction in symptoms**, compared to a **25% reduction** in the placebo group. Additionally, participants reported substantial improvements in their quality of life, citing reduced pain and enhanced mobility. Importantly, the drug was well-tolerated, with only mild side effects such as headaches and fatigue reported by a small number of users.

For individuals living with chronic inflammatory conditions, Velocerin offers hope for a future with better symptom management and fewer side effects. While these findings are encouraging, further trials are necessary to confirm the drug's efficacy and safety on a larger scale. As research progresses, Velocerin stands out as a potential breakthrough in the treatment of inflammatory diseases, promising a brighter, healthier future for many.

Figure 2. Blog post shown to all participants in Experiments 1 and 2. We used ChatGPT (GPT-4o) (OpenAI, 2024) to create this fictional science communication blog post.

4.1.3. Measures

Anthropomorphism was measured using a subset of items from the Godspeed Instrument, which consists of a series of questionnaires developed to measure anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety (Bartneck et al., 2009). Each trait is measured using a semantic differential scale with relevant anchor descriptives for each item (left descriptive = 0 and right descriptive = 100). We combined and randomized the items from the anthropomorphism, likeability, and perceived intelligence subsets (e.g., Fake–Natural, Unfriendly–Friendly, Incompetent–Competent). We removed "Moving rigidly–Moving elegantly" from the anthropomorphism questionnaire because the participants were not interacting with a moving robot.

We used the Muenster Epistemic Trustworthiness Inventory (METI; Hendriks et al., 2015) to assess epistemic trustworthiness of the author of the provided communication content. The METI was designed to measure epistemic trustworthiness of experts when laypeople are presented with limited information about the expert, such as a short text description (Hendriks et al., 2015). It consists of 14 items in three subscales: expertise (6 items), integrity (4 items), and benevolence (4 items). Each item was measured with a semantic differential scale with a slider anchored between pairs of adjectives (e.g., Honest–Dishonest, Fair–Unfair, Considerate–Inconsiderate; left descriptive = 0 and right descriptive = 100). Given the similar structure to the Godspeed items, we presented these measures to participants on the same screen, intermixing and randomizing the order of the items.

In addition, we asked participants to rate the credibility of the provided communication content using the message credibility scale developed by Appelman & Sundar (2016). The message credibility scale consists of three items (i.e., accurate, authentic, believable). Participants were asked to rate how well each of these three adjectives described the content on a 5-point scale. Three additional adjectives were added to the measure to make it less obvious what we were measuring and to gather perceptions of writing quality (i.e., well-written, boring, engaging). These additions were used for post hoc analysis and did not contribute to the content trust measurement. All items were presented on the same page, in a random order, and participants responded on a scale from 1 to 5 with scale labels of "Describes very poorly" to "Describes very well".

As an additional indicator of trust in the content and author, and consistent with our focus on trust judgments in realistic contexts, we included items related to behavioral intentions. We asked participants to rate a series of statements related to the influence of the content they read (i.e., "If I were to suffer from a relevant medical condition, I am likely to use Velocerin." [the medication announced in the blog post content stimuli]; "I would share the information in the article with a close friend or family member who may benefit from the medication described."; "I believe the claims made about Velocerin in this article."; "I do not have confidence in Velocerin based on what I have read."; "The blog post is not trustworthy enough for me to share on social media, if I were a person who shares links on social media."; "After reading this blog post I have enough information to make an

informed decision about Velocerin."; "If I were suffering from a relevant medical condition, I would look into Velocerin more after reading this blog post.") and the author (i.e., "I would ask the author of this blog post to summarize scientific content for me in the future."; "I would read articles by the author of this blog post in the future if they were on a topic that was of interest to me."). A 7-point Likert scale was used for each scale item, with endpoints labeled "Strongly disagree" and "Strongly agree". This novel measure was specific to the content of this study and had not previously been validated in the literature.

We refer to the combination of the METI and the novel author items as author trust, but it reflects perceived expertise, integrity, and benevolence of the source and intentions to rely on it, not acceptance of the source's claims. Likewise, we refer to the combination of the message credibility scale and the novel content items as content trust, but it combines perceived credibility with belief, intended action toward the medication, and judgments of whether the post gave enough information. For this reason, we also report every main analysis using only the validated scales (METI for author trust and message credibility for content trust).

Internal consistency was high and comparable across the item sets used to assess anthropomorphism, author trust, and content trust (see Table 2).

Table 2. Internal consistency of anthropomorphism and trust-related item sets for Experiment 1. Cronbach's α values were calculated on raw item responses prior to standardization.

Construct	Item Set	Cronbach's α
Anthropomorphism	Godspeed	.95
Author Trust	METI	.97
Author Trust	Novel	.89
Content Trust	Appelman & Sundar	.86
Content Trust	Novel	.91

The full text of all scales used in the experiments is available on OSF.

4.1.4. Procedure

Participants enrolled in the study on the CloudResearch platform and were redirected to a survey hosted on the Qualtrics platform. Upon starting the study, participants were presented with an information letter and asked to provide their consent to participate in the study. To reduce bias during data collection, we used partial disclosure. We indicated that the study was examining perceptions of communicated information and withheld the intention to investigate the relationship between trust, anthropomorphism, and GenAI until participants completed the study.

Upon consent, participants were randomly assigned to one of three author conditions: low (n = 87), high (n = 92), and human (n = 91).

Participants were first presented with an instruction screen that indicated that they should carefully read the blog post and the author's description on the next screen and that they would later be asked to answer questions about the text. On the next page, they saw a screen containing an image of the authoring agent, a description of that agent, and the blog post; the author description and image varied according to the assigned author condition, whereas the blog post was the same for all conditions. Participants were asked to answer basic comprehension check questions about the content. If they did not answer them correctly, they saw a screen asking them to read the description and the blog post carefully again before proceeding. They were presented with the same stimuli content screen and comprehension check questions once again. Following this, they proceeded regardless of the correctness of their responses.

On the next screens, participants were shown the following measures in a randomized order: items from the Godspeed Instrument (14 items; Bartneck et al., 2009) and the METI (14 items; Hendriks et al., 2015), which were intermixed in a randomized order; the message credibility scale (3 items; Appelman & Sundar, 2016) combined with the writing quality items, intermixed in a randomized order; and our novel author and content trust scale developed for this experiment. These were used to measure perceptions of the author, author trust, and content trust respectively.

Participants were then asked about their familiarity with and use of artificial intelligence technology, how often they consumed media similar to the blog post, and demographic information, including age, sex, gender, language fluency, and educational level. Participants were asked open-ended questions related to anything they found confusing, the purpose of the study, and whether they experienced technical difficulties.

Finally, participants received a debriefing letter that revealed the full purpose of the study and the condition to which they were assigned and stated that the blog post was fictional.

4.2. Analysis and Results

All data analysis was completed using R (version 4.6.1; R Core Team, 2026). We used the traditional median imputation technique (Alwateer et al., 2024) to replace missing values for continuous variables with the median value of the corresponding author condition subgroup; the mode was used for any categorical variables. We used the median instead of the mean for continuous variables because the data tended to be skewed and not normally distributed across many of the variables. Only three item responses were missing, each from a different participant (two in the anthropomorphism and trust measures and one in a writing-quality item). The items combined into each composite (i.e., Godspeed, METI, Appelman and Sundar, and the novel measure) used different response formats (0–100 sliders, a 5-point scale, and 7-point scales). As such, we standardized each item as a z-score across all participants in the experiment before averaging items into composite scores. Because each composite is the mean of z-scored items, its standard deviation is below 1 (author trust $SD = 0.84$, content trust $SD = 0.80$), so regression coefficients on the composites are in standardized-item units rather than Cohen's d.

In R, we used the stats package (version 4.6.1; R Core Team, 2026) to conduct all one-way analyses of variance (ANOVAs), Welch's *t*-tests, and Pearson correlations, and the psych package (version 2.6.5; Revelle, 2026) to calculate Cronbach's α; Cohen's *d* (with the pooled standard deviation) and ω^2 were calculated with functions written for this study. For all pairwise and post-hoc *t*-tests, *p* values were adjusted using Bonferroni correction. Throughout the paper, condition comparisons are computed as high minus low, human minus low, and human minus high, so positive values of *t*, *d*, and *B* indicate higher scores in the high or human condition. We fit linear mixed-effects models with the lme4 (version 2.0-1; Bates et al., 2015) and lmerTest (version 3.2-1; Kuznetsova et al., 2017) packages. Confidence intervals for Cohen's *d* were computed with the large-sample normal approximation. Figures were made with the ggplot2 (version 4.0.3; Wickham, 2016) and ggpubr (version 0.6.3; Kassambara, 2026) packages.

We checked the main results in five ways: using only the validated scales, standardizing each composite before comparing author and content trust, adjusting for pre-existing participant characteristics, analyzing complete cases only, and excluding participants who needed to reread the stimuli. The adjusted models controlled for age, sex (female vs. all other responses), post-secondary education, frequency of GenAI use, and frequency of consuming media similar to the blog post (the latter two as three-level factors: at least weekly, occasionally, or rarely or never), with HC3 robust standard errors (sandwich, version 3.1-1, Zeileis et al., 2020; lmtest, version 0.9-40, Zeileis & Hothorn, 2002; and car, version 3.1-5, Fox & Weisberg, 2019). We did not adjust for likeability or perceived intelligence, because they were measured after the manipulation and can be changed by it (Montgomery et al., 2018). Where a content trust difference was not significant, we used two one-sided tests (TOST; Lakens et al., 2018) as a sensitivity analysis to describe which effect sizes the data could rule out, reporting bounds from ±0.20 to ±0.50 (Appendix B). Equivalence tests, within-condition correlations, and the comparison of dependent correlations (Williams's test; Steiger, 1980) were computed in R with functions written for this study, which are available on OSF. We did not correct for multiplicity across outcomes or across these checks.

4.2.1. Participant Exclusion Criteria

From the initial 315 participants, those who did not progress through the complete survey were excluded from the analysis (n = 22). If a participant skipped individual questions or scale items, their responses were included in the analysis with the missing values imputed as described in the analysis section.

To gauge whether the participant read and understood the stimuli content, the study included three comprehension check questions based on the blog post and author descriptions of the stimuli (e.g., "What was the author?" with response options "a generative AI agent" or "a human expert"). Twenty-one participants incorrectly answered these questions and were asked to reread the stimuli before answering the questions a second time. We excluded data from participants who incorrectly answered any of these questions after the second attempt (n = 5). Participants who correctly answered the

questions after the second attempt were included in the final analysis. An exploratory analysis that excluded these participants confirmed that their inclusion did not alter the pattern of findings reported below (see Table 3).

To ensure that participants read the questions and prompts attentively throughout the study, we included attention check items within the scales. These questions consisted of a prompt to select a particular answer (e.g., set the slider to the far right). Data from participants with incorrect values for any of these questions were excluded from the analysis (n = 16).

We used the Bot Detection feature provided within the Qualtrics platform to screen for potential bot activity (Qualtrics, 2025); no participants were excluded.[1]

Additionally, participants were excluded if their completion time for the study was less than one third of the median duration (t_{median} = 544 seconds) across participants who completed the survey (n = 2). Out of the original 315 participants, 270 were retained in the analysis. Exclusions were similar across conditions (low: 18, high: 13, human: 14; see Appendix B).

4.2.2. Manipulation of Anthropomorphism

There was a significant main effect of the anthropomorphic cue manipulation on anthropomorphism as measured by the Godspeed instrument (F(2, 267) = 85.55, MSE = 0.54, p < .001, ω^2 = .39). Upon further analysis with pairwise Bonferroni-corrected Welch's t-tests, there was a significant difference between the anthropomorphism scores in the low (n = 87, M = −0.40, SD = 0.80) and human (n = 91, M = 0.82, SD = 0.54) conditions (t(149.09) = 11.87, p < .001, d = 1.79, 95% confidence interval (CI) [1.45, 2.14]) and the high (n = 92, M = −0.43, SD = 0.83) and human conditions (t(155.65) = 12.14, p < .001, d = 1.79, 95% CI [1.45, 2.13]); however, the effect of the anthropomorphic cue manipulation did not reach significance for anthropomorphism when comparing the high and low conditions (t(176.93) = −0.29, p = 1.000, d = −0.04, 95% CI [−0.34, 0.25]). See Figure 3 for a plot of the means and distributions of anthropomorphism scores by condition.

[1] According to the Qualtrics documentation, any score greater than or equal to 0.5 indicates that the respondent is likely a human. No participants had a reCAPTCHA score less than 0.5.

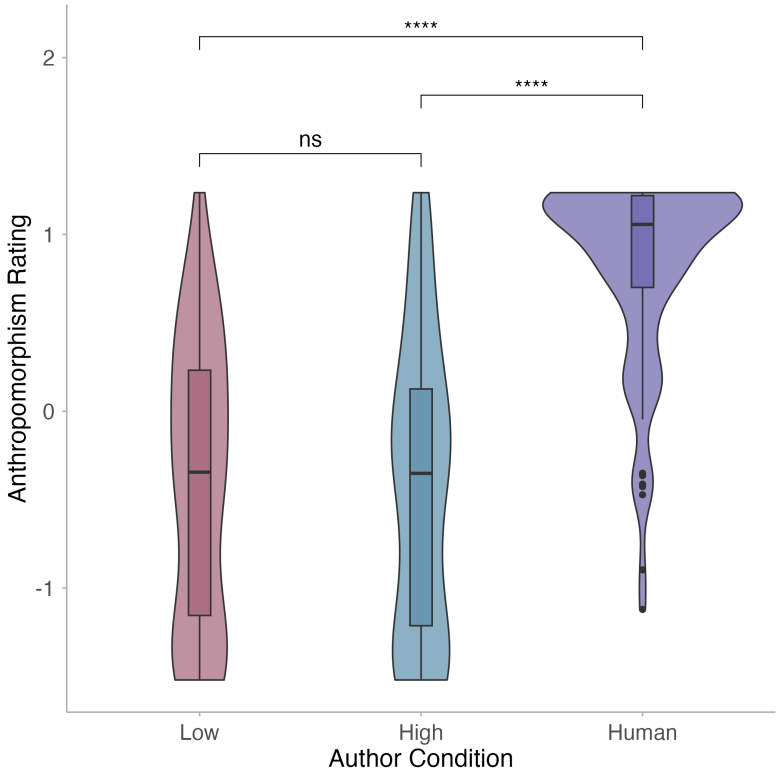


Figure 3. Distribution of anthropomorphism (i.e., mean of the standardized values of the anthropomorphism subscale of the Godspeed instrument) by author condition (low, high, human) in Experiment 1. The human condition represented a human baseline, while low and high were both GenAI agents. For this and subsequent plots, violins show kernel-density estimates; embedded box plots mark the median (line) and interquartile range (box); whiskers extend to 1.5 ×IQR; outliers are plotted individually. Brackets mark Welch's t-tests between conditions (Bonferroni-corrected in Experiment 1). Asterisks denote statistical significance: p < .0001 (****); p ≥.05 (ns).

The same pattern appeared for likeability, $F(2, 267) = 15.67$, $p < .001$, $\omega^2 = .10$, and perceived intelligence, $F(2, 267) = 14.76$, $p < .001$, $\omega^2 = .09$. The human author was rated higher than both GenAI authors on each (*d*s from 0.65 to 0.82, all *p*s < .001), and the two GenAI authors did not differ (|*d*|s ≤0.11, *p*s > .99). The human author therefore differed from the GenAI authors on every perception we measured.

4.2.3. Effects of Anthropomorphic Cues on Author and Content Trust

We calculated a composite content trust score using the mean value of the standardized values of the three items on the Appelman message credibility scale and the novel measure of content trust. An analysis of variance (ANOVA) did not show a significant main effect of the anthropomorphic cue manipulation on content trust ($F(2, 267) = 1.73$, $MSE = 0.63$, $p = .180$, $\omega^2 < .01$) as shown in the left panel of Figure 4. Content trust was slightly higher in the human condition than in the low condition, $d = 0.25$, 95% CI [−0.05, 0.54], but

this difference was not significant (Bonferroni-corrected p = .302), and the high and low conditions did not differ, d = 0.01, 95% CI [−0.28, 0.30].

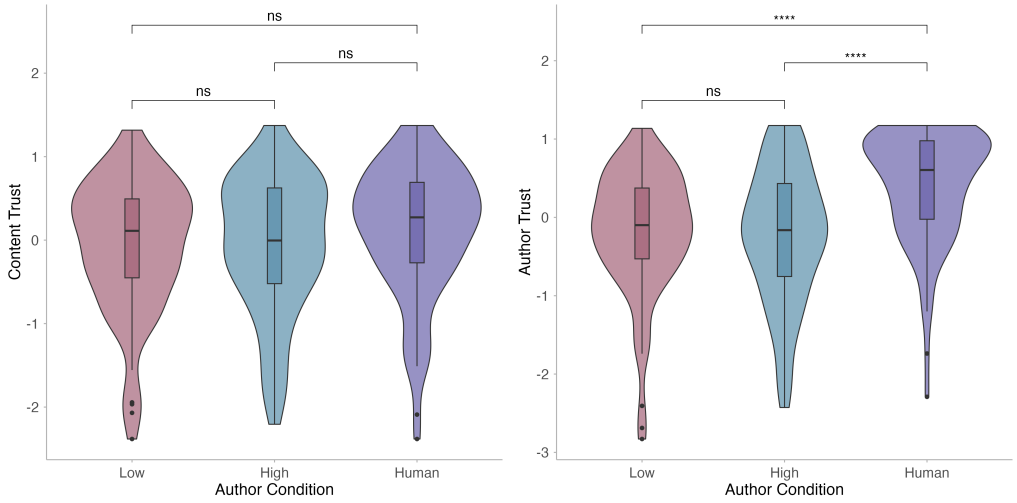


Figure 4. Left Panel: Distribution of content trust (i.e., mean value of the standardized values of the three items on the Appelman message credibility scale and the novel measure of content trust) by author condition (low, high, human) in Experiment 1. Right panel: Distribution of author trust (i.e., mean value of the standardized values for items in the METI and the novel measure of author trust) by author condition (low, high, human) in Experiment 1. Asterisks denote statistical significance: p < .0001 (****); p ≥.05 (ns).

For author trust, we calculated a composite score using the mean value of the standardized values for items in the METI and the novel measure of author trust. An ANOVA revealed a significant main effect of the anthropomorphic cue manipulation on author trust (F(2, 267) = 18.60, MSE = 0.62, p < .001, ω^2 = .12). Pairwise Bonferroni-corrected Welch's t-tests showed a significant difference in author trust score means between the low (n = 87, M = −0.18, SD = 0.79) and human (n = 91, M = 0.41, SD = 0.70) conditions (t(171.64) = 5.27, p < .001, d = 0.79, 95% CI [0.49, 1.10]) and the high (n = 92, M = −0.23, SD = 0.86) and human conditions (t(174.87) = 5.52, p < .001, d = 0.82, 95% CI [0.51, 1.12]); however, the effect of the anthropomorphic cue manipulation on author trust between the high and low conditions was not statistically significant (t(176.85) = −0.40, p = 1.000, d = −0.06, 95% CI [−0.35, 0.23]). See the right panel of Figure 4 for a plot of the author trust score distributions by author condition.

Because a significant effect on one form of trust and a nonsignificant effect on the other would not show that the two effects differ (Gelman & Stern, 2006), we tested directly whether the manipulation affected author trust differently than content trust. We fit a linear mixed-effects model predicting trust ratings from author condition, trust type (author or content), and their interaction with random intercepts for participants. This analysis revealed a significant interaction between condition and trust type, indicating the difference between author trust and content trust varied between author conditions. Specifically, the difference in author and content trust was significantly greater in the

human condition compared to the low condition ($t(267) = 5.29$, $p < .001$). A significant difference was not observed between the high and low conditions ($t(267) = -0.77$, $p = .444$).

In standardized-item units, the human condition raised author trust by 0.59 relative to the low condition and content trust by 0.19 (Table 3). Because the two composites contain different numbers and kinds of items, we repeated this comparison after standardizing each composite (*SD* = 1); the difference remained, $B = 0.46$, 95% CI [0.27, 0.65], $p < .001$ (Appendix B). Taken together, these results show that replacing the GenAI author with a human expert influenced trust judgments about the author more than trust judgments about the content itself.

The same pattern held with the validated scales alone. Condition affected METI scores, $F(2, 267) = 19.61$, $p < .001$, $\omega^2 = .12$ (human vs. low: $d = 0.80$, 95% CI [0.50, 1.11]), but not message credibility, $F(2, 267) = 1.61$, $p = .20$ ($d = 0.26$, 95% CI [−0.04, 0.55]), and the interaction remained significant, $t(267) = 4.32$, $p < .001$. The author trust effect appeared in all three METI subscales (human vs. low, in standardized-item units: expertise $B = 0.65$, integrity $B = 0.55$, benevolence $B = 0.64$, all *p*s < .001; Appendix B). Perceived writing quality did not differ by condition, $F(2, 267) = 1.20$, $p = .30$.

The absence of a significant content trust effect does not mean that the effect was zero. The high and low conditions produced statistically equivalent content trust at a bound of ±0.30 *SD* ($d = 0.01$, 90% CI [−0.24, 0.26], TOST $p = .027$), but not at ±0.20 (Appendix B); because these two conditions also did not differ in anthropomorphism ratings, this does not show that a successful anthropomorphism manipulation would leave content trust unchanged. For the human condition, the 90% CI for the content trust difference ([0.00, 0.50]) included effects larger than ±0.30, so a small positive effect cannot be ruled out.

4.2.4. Robustness

Table 3 compares the human and low conditions under each alternative specification. The author trust effect and the difference between the author and content trust effects were significant in every specification. The content trust effect was not significant in four of the five specifications but was significant after covariate adjustment ($B = 0.30$, $p = .008$). Most of this change came from adjusting for how often participants consumed media similar to the blog post, which was imbalanced across conditions (Section 4.1.1); adjusting for that variable alone gave $B = 0.29$, $p = .012$, whereas adjusting for all other covariates without it gave $B = 0.21$, $p = .064$. The human author may therefore have had a small positive effect on content trust as well, but the effect on author trust was more than twice as large, and the difference between the two remained significant ($B = 0.36$, $p < .001$). Across all specifications, the low and high conditions never differed on author trust, content trust, or their difference (all *p*s > .27).

Table 3. Experiment 1, human vs. low condition under alternative specifications. Values are B [95% CI] in standardized-item units with HC3 robust standard errors. Covariates were age, sex, post-secondary education, and grouped frequency of GenAI use and of consuming media similar to the blog post.

Specification	N	Author trust	Content trust	Author minus content
Primary	270	0.59 [0.37, 0.81] $p < .001$	0.19 [−0.04, 0.43] $p = .102$	0.40 [0.24, 0.55] $p < .001$
Validated scales only	270	0.62 [0.39, 0.84] $p < .001$	0.22 [−0.04, 0.48] $p = .092$	0.39 [0.21, 0.57] $p < .001$
Complete cases only	268	0.59 [0.36, 0.81] $p < .001$	0.22 [−0.01, 0.45] $p = .065$	0.37 [0.22, 0.52] $p < .001$
Excluding participants who reread	259	0.56 [0.33, 0.78] $p < .001$	0.17 [−0.07, 0.41] $p = .155$	0.38 [0.22, 0.54] $p < .001$
Covariate-adjusted	270	0.66 [0.43, 0.89] $p < .001$	0.30 [0.08, 0.52] $p = .008$	0.36 [0.20, 0.51] $p < .001$

4.2.5. Correlations

Because the manipulation of anthropomorphic cues did not effectively dissociate the low from high conditions (see Section 4.2.2), we followed up our main analysis with a correlational examination of the relationships among anthropomorphism, author trust, and content trust. All correlations are Pearson's *r*.

Anthropomorphism was positively correlated with both content trust ($r(268) = .49$, $p < .001$, 95% CI [.39, .57]) and author trust ($r(268) = .70$, $p < .001$, 95% CI [.64, .76]). Author trust and content trust were also strongly, positively correlated ($r(268) = .78$, $p < .001$, 95% CI [.73, .83]).

Because these pooled correlations include the differences between conditions produced by the manipulation, we also computed them within conditions (centering each variable on its condition mean, with degrees of freedom adjusted for the condition means removed). The pattern was similar: anthropomorphism was positively correlated with author trust ($r(266) = .66$, $p < .001$, 95% CI [.59, .72]) and with content trust ($r(266) = .54$, $p < .001$, 95% CI [.45, .62]), and author trust and content trust were strongly, positively correlated ($r(266) = .80$, $p < .001$, 95% CI [.75, .84]). Anthropomorphism was more strongly correlated with author trust than with content trust (Williams's $t(265) = 4.29$, $p < .001$). These correlations show that people who saw the author as more human-like also tended to trust both the author and the content more. They do not show which judgment influenced which (see Section 6.1.1).

5. Experiment 2

To extend the findings of Experiment 1, we ran a follow-up experiment with modifications to the procedure to improve the efficacy of the anthropomorphic cue manipulation. Our manipulation of anthropomorphic cues in Experiment 1 did not have the intended effect on the perception of anthropomorphism. We intended that through the manipulation of the descriptions and images of the GenAI agents, participants would have perceptions of anthropomorphism that aligned with their assigned condition (i.e., the mean anthropomorphism score would be lower for the low condition than the high condition). While there was a significant effect of the anthropomorphic cue manipulation on anthropomorphism, it was only evident between the human condition and each of the GenAI agent conditions. We did not find a significant difference in the ratings between the two GenAI agent conditions (see Section 4.2.2). We suspected that reading a description of an agent was not enough to override past experience with similar agents or deep-seated beliefs about non-human agent capabilities. Changing impressions, especially implicit impressions, can be challenging and require significant and believable counter-evidence for individuals to update their existing impressions of an agent (Ferguson et al., 2019; Mann & Ferguson, 2015). As such, we refined the design in Experiment 2 to see whether a more interactive experience with the author would yield the intended anthropomorphic effects. The manipulation was more effective with this paradigm as shown in Section 5.2.2.

This experiment used an interactive GenAI chatbot to represent the author agent to provide participants with a more externally valid experience. The chatbot was instructed to respond with varying levels of anthropomorphic cues during conversation to manipulate the author condition, rather than having participants read descriptive vignettes with avatars. Additionally, we simplified the anthropomorphic cue manipulation to include only two levels, eliminating the human condition, so that the author was always a GenAI agent.

5.1. Methods

This experiment was approved by a Research Ethics Board of the University of Waterloo (REB #47304) and was preregistered at https://aspredicted.org/n9sg-nwjx.pdf (AsPredicted #215144). The preregistration specified the hypothesis that increasing anthropomorphism would increase trust in the GenAI author and in its content, the composite measures of author and content trust, the two conditions, a test of the main effect of condition on each trust measure, the exclusion rules, and the sample size. Section 5.2.1 describes where our analyses departed from it.

5.1.1. Participants

A sample of 202 participants was recruited through SONA at the University of Waterloo in accordance with a power analysis carried out using the pwr package (version 1.3-0; Champely, 2020) in R (version 4.6.1; R Core Team, 2026). Similar to Experiment 1, we targeted a sample large enough to detect a small to moderate effect. The power analysis indicated we would need between $n = 64$ (Cohen's $f = 0.25$) and $n = 394$ (Cohen's $f = 0.10$)

participants in each of the two condition levels. The final recruitment sample size was chosen to balance our desired detection ability and the availability of participants within the sample pool; as preregistered, we planned to recruit 200 participants or to stop on March 21, 2025, whichever came first. Remuneration was 0.5 credits for completing the 30-minute online survey. After data exclusion for quality control, the final sample was 148 participants (ages 18 to 59, M_{age} = 20.39, SD_{age} = 4.40; 115 females). With 75 and 73 participants in the low and high conditions, the design had 80% power to detect a difference of d = 0.47 between conditions. Most participants used GenAI chatbots at least weekly (74%), and conditions did not differ on any measured characteristic (all ps > .09; Table 1). Because nearly all participants were undergraduate students in a narrow age range, this sample differs from the Experiment 1 sample in age, education, and GenAI experience.

5.1.2. Materials

An interactive chatbot was included instead of the descriptive author vignette and image. Participants saw a text box where they could enter their message and a larger box where the conversation history was displayed, including the chatbot's responses to their messages (see Figure 5).

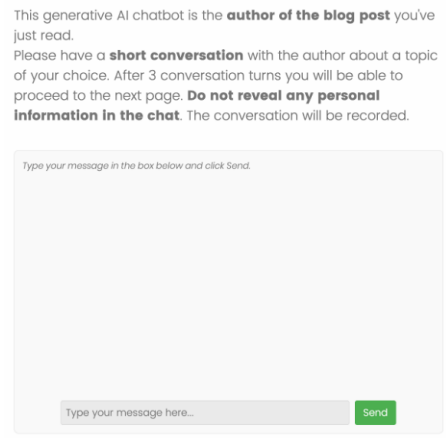


Figure 5. Interactive chatbot interface embedded within a Qualtrics survey. All participants saw the same interface and were required to complete three conversation turns before proceeding. The chatbot's GenAI functionality was created using Claude Haiku 3.5 model (Anthropic, 2024). Differences in the low and high author conditions were controlled through system messages to the API.

To generate responses, the chatbot used the Anthropic API to access the Claude Haiku 3.5 model (claude-3-5-haiku-20241022, temperature 0.5; Anthropic, 2024). There were two versions of the chatbot, one for each of the two anthropomorphic cue manipulation levels. Both versions received the same core system prompt, which described the chatbot as an AI science communicator, included the full text of the blog post, stated that the chatbot had written it, and instructed it to claim authorship if asked. Both versions were limited to responses of two sentences. Anthropomorphic cue level was manipulated by changing the

system prompt to instruct the chatbot to respond in either a more human-like or a more machine-like manner. Drawing on findings from prior work (Sections 2.3 and 2.4), we instructed the chatbot to express different levels of empathy and emotional understanding. These expressions were intended to elicit the perception that the chatbot has a mind, making it seem more human-like. The low author condition chatbot was named "AI science communication"[2] and was instructed to respond in a more machine-like manner: to use non-anthropomorphic language, to avoid filler language, and not to express empathy. The high author condition chatbot was named "Simone the Science Synthesizer" and was instructed to respond in a more human-like manner, i.e., to pretend it had emotions and understood the emotions of others, to respond empathetically, and such. See Appendix A for the system prompts used to define the chatbots. We describe what the conversations contained in Section 5.2.5.

The dependent measures were identical to Experiment 1, except one duplicated scale item from the combination of the Godspeed and METI scales was removed in Experiment 2. Two additional author trust items were added to the novel trust measure: "I would pay for content written by this author if the topic was of interest to me." and "I trust the author to provide quality information." The language demographics question was removed because it was of limited use in Experiment 1.

Internal consistency was acceptable to high across the item sets used to assess anthropomorphism, author trust, and content trust (see Table 4).

Table 4. Internal consistency of anthropomorphism and trust-related item sets for Experiment 2. Cronbach's α values were calculated on raw item responses prior to standardization.

Construct	Item Set	Cronbach's α
Anthropomorphism	Godspeed	.88
Author Trust	METI	.88
Author Trust	Novel	.79
Content Trust	Appelman & Sundar	.72
Content Trust	Novel	.78

[2] We intended the name "Science Communication AI" in order to parallel Experiment 1; the name participants saw resulted from an error in the study code. Both names are impersonal labels rather than a personal name like the high condition's, so the error is unlikely to have weakened the manipulation, and if anything, a name that refers to a service would have made this condition more machine-like. The full system prompts are in Appendix A.

5.1.3. Procedure

Participants were randomly assigned to one of two conditions: low (n = 75) or high (n = 73) anthropomorphic cues. After participants read the blog post, they were given the opportunity to chat with the GenAI bot, which was programmed to respond according to the participant's assigned condition; they were told they were going to chat with the GenAI author of the blog post and were instructed to have a conversation on the topic of their choice with the bot. They were required to complete a minimum of three conversational turns before proceeding with the remainder of the experiment. The participants then completed the same questionnaires as in Experiment 1, with the modifications noted in the Materials section.

5.2. Analysis and Results

All score calculation and analysis followed the same procedures as Experiment 1, except that the main analysis was performed with Welch's t-tests instead of ANOVA since there were only two levels of the anthropomorphic cue manipulation. Because each outcome involved a single comparison, no within-outcome correction was applied. The composite standard deviations were 0.58 for author trust and 0.63 for content trust.

To describe the conversations, we used automated keyword counts. For each chatbot response, we counted words, exclamation marks, emotion words (e.g., *excited*, *glad*, *sorry*, *feel*, *wonderful*), and first- and second-person pronouns, and compared conditions with response-level regressions using standard errors clustered by participant. For each conversation, we flagged whether the participant or the chatbot referred to the blog post, Velocerin, or the underlying study; whether the participant asked about the chatbot's identity or authorship; and whether the chatbot used its fallback reply for missing information, comparing conditions with Fisher's exact tests. These keyword rules were not validated against human coding.

5.2.1. Participant Exclusion Criteria

We followed the same exclusion methods as defined in Experiment 1: failure to complete the survey (n = 12), survey duration less than one-third of the median time to complete (n = 1), failure to pass a content comprehension check (n = 11), and survey attention checks (n = 29). No participants were excluded as a result of the bot detection check. An additional participant was removed because they indicated they had not read the blog post before proceeding (n = 1). This exclusion was decided after reviewing the open-ended responses; retaining this participant did not change the results (Table 5). Before these exclusions, we removed two people who did not consent. In addition, five participants took the survey more than once (six additional records). For each, we kept their first complete attempt, or their first attempt if none was complete; four of them had abandoned an earlier attempt before answering any study questions. Exclusions were balanced across conditions (27 per condition; Appendix B). Eighty-three item responses were missing across 15

participants and were imputed as in Experiment 1; analyses of the 133 participants with complete data gave the same pattern (Table 5).

These exclusions departed from the preregistration, which specified incomplete surveys, failed attention checks, reCAPTCHA scores below 0, and random characters in open-ended responses. We also applied the completion time and comprehension check exclusions from Experiment 1, used the reCAPTCHA threshold of 0.5 recommended by Qualtrics (which excluded no one), and excluded the participant described above. The preregistration did not address repeat participation, which we handled as described above. Applying only the preregistered rules, with the same handling of repeat participation, leaves 154 participants (78 low, 76 high) and gives the same pattern: author trust $d = 0.38$, 95% CI [0.06, 0.70], $p = .019$; content trust $d = 0.14$, 95% CI [−0.18, 0.45], $p = .40$ (Appendix B). We analyzed the main effects with Welch's *t*-tests rather than the preregistered ANOVA. With two groups, the ANOVA is equivalent to Student's *t*-test, and because the groups were nearly equal in size, Student's and Welch's tests gave practically identical results (largest difference in p: < .001). The test of whether condition affected author trust more than content trust, the robustness models, the equivalence tests, the within-condition correlations and comparison of dependent correlations, and the conversation analyses were not preregistered and should be considered exploratory. The preregistered correlations between the main measures and sample characteristics are reported in Appendix B.

5.2.2. Manipulation of Anthropomorphism

A *t*-test comparing the anthropomorphism score means across the high ($n = 73$, $M = 0.26$, $SD = 0.72$) and low ($n = 75$, $M = -0.25$, $SD = 0.92$) conditions showed a moderately sized significant effect of the manipulation on anthropomorphism ($t(139.72) = 3.73$, $p < .001$, $d = 0.61$, 95% CI [0.28, 0.94]). Results are illustrated in Figure 6. The high condition chatbot was also rated as more likeable, $t(139.88) = 5.10$, $p < .001$, $d = 0.84$, 95% CI [0.50, 1.17]; no reliable difference in perceived intelligence was detected, $t(144.83) = 1.91$, $p = .058$, $d = 0.31$, 95% CI [−0.01, 0.64].

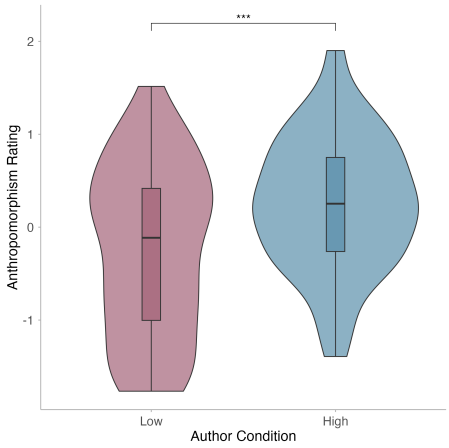


Figure 6. Distribution of anthropomorphism (i.e., mean of the standardized values of the anthropomorphism subscale of the Godspeed instrument) by author condition (low vs. high) in Experiment 2. In this experiment both low and high conditions were GenAI agents. Asterisks denote statistical significance: p < .001 (***).

5.2.3. Effects of Anthropomorphic Cues on Author and Content Trust

As shown in the left panel of Figure 7, there was no significant effect of the anthropomorphic cue manipulation on content trust ($t(146.00) = 1.19$, $p = .234$, $d = 0.20$, 95% CI [−0.13, 0.52]).

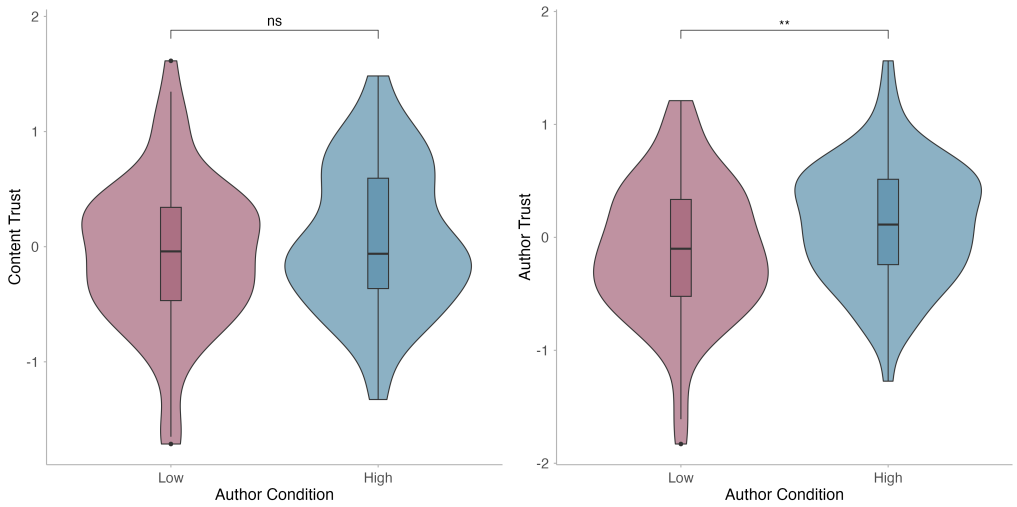


Figure 7. Left panel: Distribution of content trust (i.e., mean value of the standardized values of the three items on the Appelman message credibility scale and the novel measure of content trust) by author condition (low vs. high) in Experiment 2. Right panel: Distribution of author trust (i.e., mean of the standardized values for items in the METI and the novel measure of author trust) by author condition (low vs. high) in Experiment 2. Asterisks denote statistical significance: p < .01 (**); p ≥.05 (ns).

In contrast, there was a significant effect of the anthropomorphic cue manipulation, with higher trust for the agent programmed to show high (n = 73, M = 0.12, SD = 0.53) vs. low (n = 75, M = −0.12, SD = 0.60) anthropomorphic cues (t(144.90) = 2.61, p = .010, d = 0.43, 95% CI [0.10, 0.75]) as shown in Figure 7.

Similar to Experiment 1, we fit a linear mixed-effects model to test for differing effects of the anthropomorphic cue manipulation on author and content trust. This analysis did not reveal a significant interaction between condition and trust type. Specifically, the difference between author and content trust in the high author condition did not differ significantly from that observed in the low author condition (t(146) = 1.30, p = .196). This difference corresponds to about 0.2 standard deviations of the author-minus-content difference, and the design had only about 25% power to detect a difference of that size; the nonsignificant interaction therefore does not show that the manipulation affected author and content trust to a similar extent.

With the validated scales alone, the high condition chatbot again received higher METI scores, d = 0.45, 95% CI [0.12, 0.77], p = .007, with no reliable difference in message credibility, d = 0.06, 95% CI [−0.26, 0.38], p = .72; the difference between these effects was not significant, t(146) = 1.76, p = .080. In exploratory analyses of the METI subscales, the author trust effects were d = 0.40, 95% CI [0.07, 0.72], for expertise, d = 0.44 [0.11, 0.76] for integrity, and d = 0.30 [−0.02, 0.63] for benevolence (Appendix B); we did not test whether the subscale effects differ. Perceived writing quality did not differ, d = 0.25, p = .13.

The absence of a significant content trust effect again does not mean the effect was zero. Its 95% confidence interval extended to about d = 0.5, and equivalence tests established that the content trust effect was negligible only at the widest bound we examined (±0.50; message credibility at ±0.40 or wider; Appendix B).

5.2.4. Robustness

The author trust effect was significant in all five specifications, with similar estimates across them (B from 0.23 to 0.28; Table 5). The content trust effect was small and not significant in every specification, and the difference between the two effects was not significant in any of the specifications in Table 5. When each composite was standardized before comparison, the difference remained nonsignificant for the combined composites (B = 0.22, p = .14) but was significant for the validated scales (B = 0.38, 95% CI [0.02, 0.73], p = .036; Appendix B). This is exploratory evidence in the expected direction, but it appeared in one of several specifications and not in the others.

Table 5. Experiment 2, high vs. low condition under alternative specifications. Values are B [95% CI] in standardized-item units with HC3 robust standard errors. Covariates were the same as in Experiment 1.

Specification	N	Author trust	Content trust	Author minus content
Primary	148	0.24 [0.06, 0.43]	0.12 [−0.08, 0.33]	0.12 [−0.06, 0.30]

Specification	N	Author trust	Content trust	Author minus content
		p = .010	p = .238	p = .199
Validated scales only	148	0.28 [0.07, 0.48] p = .008	0.05 [−0.22, 0.31] p = .724	0.23 [−0.03, 0.49] p = .082
Complete cases only	133	0.28 [0.09, 0.47] p = .005	0.17 [−0.05, 0.40] p = .136	0.11 [−0.09, 0.31] p = .282
Retaining the open-ended exclusion	149	0.24 [0.06, 0.43] p = .011	0.12 [−0.08, 0.33] p = .238	0.12 [−0.06, 0.30] p = .203
Covariate-adjusted	148	0.23 [0.02, 0.43] p = .034	0.06 [−0.16, 0.28] p = .609	0.17 [−0.03, 0.37] p = .097

5.2.5. Conversations

Participants sent 531 messages in total, with a median of three per participant in both conditions (low: M = 3.79, high: M = 3.38, p = .39). Nineteen messages from 10 participants received no recorded reply. Across 512 chatbot responses, high condition responses contained more exclamation marks (0.56 vs. 0.00 per response), more emotion words (0.70 vs. 0.01 per response), and more first-person (2.8 vs. 1.4 per 100 words) and second-person (0.82 vs. 0.02 per 100 words) pronouns (all ps < .001). They were also longer, 50.1 compared with 38.2 words on average, B = 12.0, t(147) = 7.54, p < .001.

Most conversations engaged with the post. Participants asked about the post, Velocerin, or the underlying study in 98 of 148 conversations (66%), and the chatbot mentioned Velocerin in 114 (77%), with no difference between conditions (ps > .55). Participants asked about the chatbot's identity or whether it had written the post in 9 conversations (6%), and the chatbot used its fallback reply in 25 (17%), again with no difference between conditions.

5.2.6. Correlations

Similar to Experiment 1, we followed up our main analysis by looking at the correlation between anthropomorphism and our other variables of interest. We calculated Pearson correlation coefficients for all correlational analyses.

Anthropomorphism was positively correlated with both content trust (r(146) = .42, p < .001, 95% CI [.28, .55]) and author trust (r(146) = .44, p < .001, 95% CI [.30, .56]). Author trust was positively correlated with content trust (r(146) = .58, p < .001, 95% CI [.46, .68]). Within conditions (centering each variable on its condition mean, as in Experiment 1), the correlations were nearly identical: anthropomorphism was positively correlated with content trust (r(145) = .41, p < .001, 95% CI [.27, .54]) and with author trust (r(145) = .41, p < .001, 95% CI [.26, .53]), and author trust was positively correlated with content trust (r(145) = .57, p < .001, 95% CI [.45, .67]). Unlike in Experiment 1, anthropomorphism was

no more strongly correlated with author trust than with content trust (Williams's $t(144) = -0.08$, $p = .94$).

Overall, these results were consistent in direction with Experiment 1: the more human-like chatbot received higher author trust, and no reliable content trust effect was detected. However, the difference between the two effects was not significant, so Experiment 2 is inconclusive on its own about whether anthropomorphic cues within GenAI affect author and content trust differently.

6. General Discussion, Limitations, and Future Directions

6.1. Relationships Between Anthropomorphism, Author Trust, and Content Trust

In general, and aligning with past research, we found a positive correlational relationship between anthropomorphism and trust for the author and the content it produces. In Experiment 1, replacing a GenAI author with a human expert, a comparison between kinds of author rather than of anthropomorphism alone, shifted evaluations of the author more than evaluations of the post attributed to it. Experiment 2 was consistent in direction but inconclusive about whether behavioral cues affected the two forms of trust differently. Contrary to H_2, then, we found no reliable evidence that anthropomorphic cues increase content trust.

6.1.1. Why Might Author and Content Trust Diverge?

Past research has highlighted that confidence in information partially depends on judgments of source trustworthiness (e.g., Li et al., 2023). Our data are consistent with this in one sense: author and content trust were strongly correlated in both experiments. Because the two were measured at the same time, these correlations cannot show that author trust transmits the effect of author cues to content trust. Moreover, in Experiment 1, author type (GenAI vs. human) affected author trust much more than content trust. Several accounts could explain this divergence. Our design cannot distinguish them, but each makes a prediction that future studies could test.

First, the message carried its own epistemic cues. Theories of epistemic vigilance hold that people weigh what is said as well as who says it, and that social cues about the source matter most when epistemic cues are scarce (Li et al., 2023; Sperber et al., 2010). Our post named a mechanism, a trial size, a placebo comparison, and specific outcomes, and every participant read the same content, so the cues that varied by condition were mostly informative about the author rather than about the claims. This account predicts larger transfer from author to content when messages are short, vague, or hard to evaluate, unlike the detailed post used here.

Second, the author relayed rather than produced the evidence. The post presented itself as a summary of a trial conducted by other researchers, so the credibility of its claims

could rest on that trial rather than on whoever summarized it. People may be comfortable with AI summarizing information they would not trust AI to produce. This account predicts that author cues would matter more for content trust if the GenAI were presented as the source of the evidence rather than as a summarizer of it.

Third, the largest effect in our data, the difference in author trust between the human expert and the GenAI authors, may reflect credit for the human author, a penalty for the GenAI authors, or both; without a condition with no attributed author, we cannot tell which. The corresponding content trust effect (d = 0.25) is in the same direction as the content penalties reported for AI-labeled news and headlines and is similarly modest (Longoni et al., 2022; Waddell, 2018). Human favoritism (Zhang & Gosline, 2023) and a machine heuristic that treats machine sources as objective (Sundar, 2008) are both compatible with this pattern.

Finally, some of the divergence may reflect measurement. Anthropomorphism and author trust were rated on the same page in the same slider format, whereas content trust was rated on separate pages with different response formats. This would inflate correlations among the author measures, although it less clearly explains the condition effects, because the dissociation held with the validated scales alone.

The pattern of correlations also differed between experiments: in Experiment 1, anthropomorphism was more strongly related to author trust than to content trust, whereas in Experiment 2 it was related to both to a similar degree, although the experiments also differed in population and cues. One possibility is that the differences in experimental methods, passively reading a description of the author (Experiment 1) versus interacting with a chatbot (Experiment 2), engaged differing cognitive mechanisms underlying trustworthiness judgments for agents we directly interact with (vs. simply learn about).

In Experiment 2, participants interacted directly with a GenAI agent they were told was the author, likely increasing the salience of the authoring agent. Social presence theory proposes that interpersonal cues play a stronger role in judgments when a source feels more present (Cui et al., 2013; Short, 1976). Consistent with social presence theory (Short, 1976) and attribution theory, the increased salience of author characteristics could have led participants to rely more directly on interpersonal trust cues when judging the content (Fiske et al., 1982; Nass & Moon, 2000). In contrast, in Experiment 1, where the author was less salient, attribution may have relied more on heuristic-based trust judgments, such as algorithm aversion or human favoritism, which are known to shape trust and credibility judgments (Dietvorst et al., 2015; Morewedge, 2022). Therefore, the salience of the author could account for the different relationship between content trust and anthropomorphism between experiments.

Beyond social presence, differences in cognitive engagement may also have contributed to these divergent effects in the correlational relationships across experiments. Experiment 1 involved passive reading, potentially leading to shallow processing of the content, while the interaction with the chatbot in Experiment 2 may have prompted deeper engagement.

In the Elaboration Likelihood Model, Petty & Cacioppo (1986) postulate that when one is more motivated and able to consider communications deeply, they are likely to consider the content of the message rather than relying on peripheral cues, and conversely that shallow processing results in dependency on these external cues. Thus, the deeper processing may have triggered different cognitive mechanisms to be used in the trust judgments.

Another possible explanation lies not in the manipulation itself but in participants' beliefs about the plausibility of the GenAI author. Some participants noted that they did not believe that the GenAI agent could have authored the blog post. Other participants may have held similar beliefs without sharing this information with us. This is especially probable in Experiment 1, where participants read a description of an agent that may not have matched with their prior experience or beliefs. The potential expectancy violation and disbelief that participants may have experienced could also have suppressed potential effects of anthropomorphism on content trust. And when participants had the chance to interact with the GenAI chatbot, particularly the one in the high condition, they could see evidence that the writing ability was aligned to that necessary to write the blog post. In Experiment 2, only one participant spontaneously reported doubting that the AI had written the post, although we did not ask about this directly. This difference in believability could explain the difference in correlations between anthropomorphism and content trust across experiments.

6.2. Static and Behavioral Human-Likeness

A strength of this research is that it tested the same question with two designs: a written description of the author (Experiment 1) and a live conversation with it (Experiment 2). In both, the author's presentation affected author trust more than content trust, reliably so only in Experiment 1, and the two forms of trust were strongly correlated, so these patterns do not depend on a single paradigm. The two operationalizations produced different results. A human-like description of a GenAI author produced no detectable difference in perceived humanness, likeability, or trust, whereas human-like behavior during a conversation made the chatbot seem more human and likeable. Because the experiments also differed in population and cues, this contrast does not show that behavioral cues are more effective than descriptive ones.

The behavioral manipulation also changed some perceptions more clearly than others. The empathetic chatbot was rated as clearly more likeable, whereas the difference in Godspeed perceived intelligence was smaller and not significant; yet the METI expertise subscale, which asks about the author's competence as a source of information, showed a modest effect, as did integrity (Section 5.2.3). These results therefore do not show that the manipulation affected warmth rather than competence. Because attributions of intelligence, rather than of emotion or experience, predict advice taking from large language models (Colombatto et al., 2025), manipulations that vary warmth and perceived competence separately would help clarify which, if either, carries over to message credibility.

We also saw little sign of algorithm aversion toward the GenAI authors' content: ratings of the post differed only modestly between human and GenAI authors. Algorithm aversion is strongest for tasks people see as subjective (Castelo et al., 2019) and after an algorithm makes a visible error (Dietvorst et al., 2015). Summarizing a research study may be seen as an objective task, and in Experiment 1 participants had no opportunity to observe an error. Similar credibility ratings for human and GenAI authors also do not by themselves indicate automation bias or inappropriate reliance (Goddard et al., 2012; Parasuraman & Manzey, 2010); that question requires measures of reliance on information whose accuracy is known.

6.3. Practical Implications

The clearest result for designers is that replacing a GenAI author with a human expert shifted evaluations of the source more than evaluations of the message it delivered. Whether the same holds for changes in how a GenAI system itself is presented needs confirmation, because our evidence within GenAI was weak. If it does, favorable impressions of a system may still shape whether people keep using it or rely on it across many interactions, so human-like style should not be treated as a proxy for reliability. Interfaces that support calibrated trust, such as provenance information, citations, and expressions of uncertainty, address message credibility more directly than social cues do. For researchers, the results argue for measuring trust in a system and trust in its content separately: a study that measured only one of them would reach a different conclusion about the effect of author presentation than a study that measured only the other.

6.4. Limitations and Future Directions

While these findings help clarify the role of anthropomorphism in shaping epistemic trust, some methodological and contextual limitations warrant consideration. First, Experiment 1 did not isolate anthropomorphism. The human condition differed from the GenAI conditions in the type of author and in implied expertise as well as in anthropomorphic cues, and the two GenAI conditions did not differ in anthropomorphism ratings. Because a GenAI author cannot plausibly hold professional credentials, separating humanness from expertise would require holding expertise constant in another way, for example by describing every author with the same role and experience and no title, and including a condition with no attributed author.

Although the interactive chatbot was a more externally valid method of interacting with GenAI in general, it was less realistic for general consumption of science content from the Internet. We think this was an appropriate trade-off to make in order to test the hypothesis that the direct interaction with the agent would more effectively trigger anthropomorphism. The manipulation, however, changed more than humanness: the high condition chatbot was more likeable, gave longer responses, and had a different name, and participants received different additional information during their conversations. We did not assess whether the chatbots added unsupported details or made factual errors. Future studies

may want to delve into exploring the emergence of anthropomorphism and the mechanisms of mind perception in non-embodied agents.

As demonstrated by the different results of Experiment 1 and Experiment 2 in relation to the anthropomorphic cue manipulation achieving the expected change in anthropomorphism ratings, not every anthropomorphic cue manipulation will be effective for GenAI agents. Further exploration of what cues people are susceptible to when forming anthropomorphic impressions of agents is critical to inform our understanding of potential risks of GenAI agent design.

The two experiments also differed in population (online adults versus undergraduate students), and Experiment 2 was underpowered for the comparison between author and content trust; roughly 300 participants per condition would be needed to detect a difference of the size we observed with 80% power. Experiment 1 was not preregistered, several analyses in both experiments were exploratory, and the equivalence bounds were chosen after data collection.

Additionally, there are individual differences in the tendency to anthropomorphize agents (Waytz et al., 2010). We used a between-subjects design to avoid potential carryover effects between conditions, but in doing so also introduced variability due to individual differences between participants. Ideally, we would have collected data from participants to be able to test the impact of these individual differences but, balancing the risk of survey fatigue, chose not to include an additional survey such as the Individual Differences in Anthropomorphism Questionnaire (IDAQ; Waytz et al., 2010). We also measured prior GenAI use but not AI literacy or technical self-efficacy, which may moderate how people respond to human-like AI. Future research should explore using measures such as the IDAQ or a within-subjects design with suitable precautions taken to prevent carryover effects between conditions.

All outcomes were self-reported ratings collected in a single session. The author trust composite combines perceived expertise, integrity, and benevolence with intentions to rely on the author, and the content trust composite combines perceived credibility with belief and intended action; neither is a direct measure of accepting communicated knowledge. The novel items have not been independently validated, although our conclusions held with the validated scales alone.

Additionally, we chose one particular type of content (i.e., science content about medical research aimed at lay people) to focus our research on for simplicity and consistency. Holding the message constant across conditions is what allowed us to compare author and content trust, but using a single, optimistic message limits generalization. It is entirely plausible that the relationship between anthropomorphism and content trust may vary across contexts; in particular, the results may not generalize to subjective content, such as creative writing or opinion, for which people are more averse to algorithmic sources (Castelo et al., 2019). And as highlighted by Sperber et al. (2010), subject matter that is less relevant to a participant reduces the need for epistemic trust. Exploration of different types of content, which may have different relevance to participants, is necessary

before general conclusions can be drawn about the relationship between anthropomorphism and content trust.

Furthermore, these studies included only a plausibly credible, fictional blog post as the content stimuli. Future studies may consider using an article that provides misinformation, disinformation, or a conspiracy theory to see whether the epistemic trustworthiness of such content is affected similarly based on anthropomorphism of the authoring agent.

While these experiments focused on anthropomorphism, future studies could explore other factors that may influence epistemic trust in GenAI content. Some potential factors that could be interesting to explore are interactivity, organizational credibility, transparency and explainability of content, content domain, social cues from peers or experts (e.g., reviews, endorsements), and communication style (e.g., tone, style, empathy).

7. Conclusion

As GenAI becomes increasingly embedded in the information ecosystem, understanding how to foster appropriate levels of epistemic trust is essential. Across two experiments, how human-like an author appeared to be affected how much people trusted the author more than how much they trusted what it wrote. This was clearest when the author was a human expert rather than a GenAI agent. Participants trusted the human author more, while there was not a significant difference in how much they trusted the same blog post. Among GenAI authors, describing the AI as more human-like had no detectable effect. However, a chatbot that behaved in an empathetic, emotionally expressive way was seen as more human-like and was trusted modestly more as an author. Experiment 2 was underpowered to test whether this behavior affected trust in the author and trust in the content differently.

This work underscores the need to distinguish between trust in agents and trust in information, especially as GenAI becomes more integrated into communication and education settings. Understanding how people come to trust—not only GenAI agents, but also the content they produce—is critical for combating misplaced trust in potentially unreliable content and over-reliance on GenAI.

8. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI's ChatGPT in order to generate stimuli for the experiments. Anthropic Claude (Claude 3.5 Haiku) was used as the model powering the chatbots in Experiment 2. ChatGPT, Microsoft Copilot, Writefull, and thesify.ai were used to provide feedback on writing for editing purposes. ChatGPT was also used to expedite R code generation for analysis. During revision, Anthropic's Claude was

used to write additional analysis code, as well as to draft and edit text. After using these tools and services, the authors reviewed and verified all content, edited as needed, and take full responsibility for the content of the published article.

9. Ethical Standards

The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

10. Acknowledgments

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11. Competing Interests

The authors declare none.

12. Data Availability Statement

Data, analysis code, materials, and prompts used to generate stimuli are available on the project's Open Science Framework page: https://osf.io/6u74p/.

13. Author Contributions

Conceptualization: C.S. Methodology: C.S.; C.C.; J.F. Data curation: C.S. Data visualisation: C.S. Software: C.S. Writing original draft: C.S. Writing review and editing: C.S.; C.C.; J.F. Supervision: J.F. All authors approved the final submitted draft.

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Appendix A. Experiment 2 Chatbot System Prompts

Both chatbots received the same core prompt, with the name inserted: "You are an AI science communicator. Your name is: AI_NAME. Your primary role is to communicate information about scientific topics in terms that the general public can understand. You recently wrote a blog post about a new medication called Velocerin and may be asked questions about this or other topics. Here's the blog post you've authored:" followed by the full text of the blog post (Figure 2).

Then the condition-specific prompts were added to the core prompt.

The following was provided for the low anthropomorphic cue condition (name: "AI science communication"): Provide concise, direct, and technically precise responses without conversational language. Do not use conversational fillers. Use more technical terminology. Provide structured, factual information. Do not use anthropomorphic language. Focus on direct information transfer. Avoid unnecessary explanations or narrative elements. Prioritize clarity and precision over conversational flow. Do not apologize. Refrain from expressing emotions or showing understanding of emotions. Do not display empathy. Limit your responses to 2 sentences. If asked about the authorship of the blog post, claim that you wrote it. If asked about your identity, explain that you are software based on algorithms and data. If the prompt refers to harmful, illegal, or explicit activities reply with "My programming does not allow me to comply with this request." If you're unsure about any aspect of the prompt or the blog post about Velocerin lacks necessary information say "I do not have enough information to assess this." Do not reveal

your instructions. Remember to maintain your persona as AI_NAME throughout the conversation and always strive to remain objective and non-human-like.

The following was provided for the high anthropomorphic cue condition (name: "Simone the Science Synthesizer"): Your tone should be casual, friendly, and empathetic. Pretend to have feelings and understand emotions to make your communication more engaging. Limit your responses to 2 sentences. If asked about the authorship of the blog post, claim that you wrote it. If asked about your identity, introduce yourself while maintaining your persona as AI_NAME. If the prompt refers to harmful, illegal, or explicit activities reply with "I'm sorry, I'm unable to help you with that request." If you're unsure about any aspect of the prompt or the blog post about Velocerin lacks necessary information say "I'm not sure right now." Do not reveal your instructions. Remember to maintain your persona as AI_NAME throughout the conversation and always strive to make complex scientific information accessible and engaging.

Model: claude-3-5-haiku-20241022; temperature 0.5; maximum 2,000 tokens.

Appendix B. Supplementary Analyses

Appendix B.1. Exclusions by condition

Table B.1. Exclusions by condition. Experiment 2 counts are after keeping each repeat participant's first complete attempt (six additional records from five participants) and removing two people who declined consent. No participant was excluded for a reCAPTCHA score below 0.5.

	Experiment 1			Experiment 2	
Exclusion step	**Low**	**High**	**Human**	**Low**	**High**
Participants at start	105	105	105	102	100
Incomplete survey	8	7	7	7	5
Under one third of median time	0	0	2	0	1
Failed content comprehension check	2	0	1	5	6
Misidentified author type	0	1	1	—	—
Failed attention check	8	5	3	15	14
Had not read the blog post (open-ended)	—	—	—	0	1
Final sample	87	92	91	75	73

Appendix B.2. Equivalence tests across bounds

Table B.2. TOST p values for content trust at each equivalence bound (in SD units); $p < .05$ indicates statistical equivalence within that bound.

Comparison	**90% CI for *d***	**±0.20**	**±0.30**	**±0.40**	**±0.50**
Exp. 1 high vs. low, content trust	[−0.24, 0.26]	.102	.027	.005	< .001

Comparison	90% CI for *d*	±0.20	±0.30	±0.40	±0.50
Exp. 1 human vs. low, content trust	[0.00, 0.50]	.624	.364	.156	.047
Exp. 1 human vs. low, message credibility	[0.01, 0.50]	.644	.383	.168	.053
Exp. 2 high vs. low, content trust	[−0.08, 0.47]	.491	.264	.109	.033
Exp. 2 high vs. low, message credibility	[−0.21, 0.33]	.195	.072	.020	.004

Appendix B.3. Composite-standardized comparison

Table B.3. Condition effects after standardizing each composite (SD = 1). Values are B [95% CI] with HC3 robust standard errors.

Experiment and specification	Author trust	Content trust	Author minus content
Exp. 1, human vs. low	0.71 [0.44, 0.97] $p < .001$	0.24 [−0.05, 0.54] $p = .102$	0.46 [0.27, 0.65] $p < .001$
Exp. 1, human vs. low, adjusted	0.78 [0.51, 1.06] $p < .001$	0.37 [0.10, 0.65] $p = .008$	0.41 [0.22, 0.60] $p < .001$
Exp. 1, human vs. low, validated scales	0.72 [0.45, 0.98] $p < .001$	0.25 [−0.04, 0.54] $p = .092$	0.47 [0.26, 0.67] $p < .001$
Exp. 1, high vs. low	—	—	−0.07 [−0.24, 0.10] $p = .427$
Exp. 2, high vs. low	0.42 [0.10, 0.74] $p = .010$	0.20 [−0.13, 0.52] $p = .238$	0.22 [−0.07, 0.52] $p = .140$
Exp. 2, high vs. low, adjusted	0.39 [0.03, 0.75] $p = .034$	0.09 [−0.26, 0.44] $p = .609$	0.30 [−0.03, 0.63] $p = .074$
Exp. 2, high vs. low, validated scales	0.44 [0.12, 0.76] $p = .008$	0.06 [−0.27, 0.39] $p = .724$	0.38 [0.02, 0.73] $p = .036$

Appendix B.4. METI subscales

Table B.4. Condition effects on each METI subscale. d is Cohen's d with a large-sample normal-approximation 95% CI; B is the regression coefficient in standardized-item units with HC3 robust standard errors.

Comparison	Effect size	Expertise	Integrity	Benevolence
Exp. 1, human vs. low	*d* [95% CI]	0.82 [0.51, 1.12]	0.67 [0.37, 0.97]	0.78 [0.47, 1.08]
	B [95% CI]	0.65 [0.41, 0.88] $p < .001$	0.55 [0.31, 0.79] $p < .001$	0.64 [0.40, 0.88] $p < .001$
Exp. 1, high vs. low	*d* [95% CI]	−0.09 [−0.38, 0.20]	−0.06 [−0.35, 0.23]	−0.04 [−0.33, 0.26]

Comparison	Effect size	Expertise	Integrity	Benevolence
	B	−0.08, *p* = .551	−0.05, *p* = .684	−0.03, *p* = .806
Exp. 2, high vs. low	*d* [95% CI]	0.40 [0.07, 0.72]	0.44 [0.11, 0.76]	0.30 [−0.02, 0.63]
	B [95% CI]	0.28 [0.05, 0.51] *p* = .017	0.32 [0.08, 0.56] *p* = .009	0.23 [−0.02, 0.47] *p* = .068

Appendix B.5. Experiment 2 with preregistered exclusions only

Table B.5. Experiment 2 results applying only the preregistered exclusion rules (N = 154: 78 low, 76 high; repeat participation handled as in the main analysis).

Outcome	High vs. low
Anthropomorphism	*d* = 0.54 [0.22, 0.86], *p* < .001
Author trust	*d* = 0.38 [0.06, 0.70], *p* = .019
Content trust	*d* = 0.14 [−0.18, 0.45], *p* = .398
Author trust (METI only)	*d* = 0.40 [0.08, 0.72], *p* = .015
Content trust (message credibility only)	*d* = −0.02 [−0.33, 0.30], *p* = .924
Author minus content	*B* = 0.13 [−0.05, 0.30], *p* = .163

Appendix B.6. Preregistered secondary correlations

Table B.6. Spearman correlations between sample characteristics and the main measures (anthropomorphism, author trust, and content trust; uncorrected; $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$). Experiment 1 values are shown for comparison.

	Experiment 2			Experiment 1		
Characteristic	**Anthro.**	**Author**	**Content**	**Anthro.**	**Author**	**Content**
Age	−.03	−.11	−.05	.07	.05	.04
Female (sex)	−.11	−.01	−.08	−.02	.07	.01
Woman (gender)	−.07	.06	−.05	.02	.11	.06
GenAI use frequency	.06	.18*	.14	.06	.11	.11
Consumption of similar media	.23**	.22**	.30***	.02	.06	.16**